

2025 Conservation Biology Student Grant Proposal Assignment

Modeled after the AZA Conservation Grants Fund (CGF)

Title of Project: Urban Pollinator Habitat Restoration

Student Name: [REDACTED]

Institution: University of Nebraska at Omaha

Project Duration: February 2026 – November 2026

Project Category:

- ☒ Field Conservation
- ☐ Animal/Population Management
- ☒ Conservation Education/Engagement
- ☐ Research

Project Abstract (limit 250):

Due to rapid urbanization in natural prairie ecosystems, as well as the effects of climate change, prairie plant and pollinator populations are experiencing significant habitat loss and fragmentation (Smith, 2025). Urban areas are concrete heavy, and the minimal plants that are planted in these areas are typically ornamentals that are non-native. Sustainably minded cities are now focusing on creating greenspaces and maximizing natural environments in a concrete, heavy area. One highly beneficial way to do this is through planting native plants, in the great plains region; specifically, prairie plants (Sexton, 2020). Bringing prairie plants back into urban areas will provide habitats for plenty of native pollinator species like insects, birds, and mammals. The purpose of this project is to restore urban pollinator habitats and increase species abundance in our local area of Omaha through the use of native prairie plants. This project is on a ten-month timeline that coordinates with Nebraska's growing seasons and focuses on in situ field conservation and conservation education/engagement. Methods in the field include site assessments, habitat preparation and planting of native species seeds, ongoing maintenance, and monitoring of pollinator activity. Methods for education include community outreach programs at public libraries, schools, nature reserves, and the zoo and botanical garden. Pollinator's abundance will be consistently monitored, and information will be publicly shared.

Project Narrative:

1. Project Overview & Significance What conservation issue or biological question does your project address? Why is it important for conservation or management?

This project addresses the issue of pollinator habitat loss in urban areas with rapid development. It is critical for conservation to maintain prairie ecosystems and their respective pollinator species that rely on them and vice versa.

2. Goals, Objectives, and Actions Identify your long-term goal, immediate objectives, and the key actions you will take to meet those objectives.

My long-term goal is to restore and increase local prairie areas and pollinator species abundance as well as spread awareness and educate the public about pollinator health and pocket prairies. The key actions I will take are to establish new pocket prairies in our area by replacing ornamentals (when appropriate in partnering sites) as well as introducing new green spaces. Community outreach and workshops are also critical to this project and will be done in partnership with conservation groups like the zoo, botanical garden, local nature conservancies like Fontanelle Forest and other public spaces.

3. Methods Describe how you will carry out your project.

I will first conduct initial site assessments looking for soil quality, invasive species presence, and current pollinator activity. The areas will be prepared and then planted with native prairie forbs and grasses using plugs in spring and dormant seeding in fall. Monitoring transects will be set up to monitor plant establishment and pollinator abundance. I will partner with local nature conservancies, libraries, the zoo and botanical garden to host community workshops on native plant gardening, pollinator biology, and habitat care. Mixed seed packs will also be given out. I will coordinate volunteer plant days for people to volunteer and help plant in the local area. I will also provide ongoing maintenance of the areas including weeding, watering and continued planting. Data collected will be compiled, (I will compare pre- and post-restoration pollinator abundance using simple transect counts) and made into a report for result analysis.

4. Timeline Present a short timeline or list of milestones covering your one-year project duration.

February: Project planning, permit approvals, and partner coordination. Mar - April: Baseline surveys of site conditions and early-season pollinator presence as well as area prep like invasive plant removal and soil preparation. First community workshop. May - June: Install native plant plugs and grasses, start pollinator monitoring biweekly, and volunteer planting days and workshops. July - Aug: Ongoing maintenance, pollinator walks with community. Sept - Oct: Dormant seeding of late-season native species, installing educational signage. Nov: Data analysis and preparing final report.

5. Evaluation & Outcomes How will you determine if the project met its objectives? Include measurable indicators of success (quantitative or qualitative).

I can determine if the project met its objectives through increased pollinator abundance and plant establishment, number of community members volunteering and participating in workshops, and reduction in invasive species presence. I would specifically like to see a 20% increase in pollinator abundance when compared to pre planting counts. Also the volunteer hours recorded, number of seed packs taken and just general feedback from partners and the community.

6. Dissemination How will you share your results or communicate your impact (e.g., outreach, poster, report, social media, or collaboration with a zoo or NGO)?

The results will be presented at a FUSE style project fair and shared with the public through social media posts. The results will also be presented to community partners like the zoo for them and general public to see in person. The data will also be submitted to citizen science platforms like iNaturalist.

7. Ethical and Welfare Considerations N/A

Budget:

Budget Item:	Cost Estimate:	Description:
Travel	\$ 800	Fuel for travel to public places like the zoo, vehicle rental for equipment.
Supplies	\$ 2000	Field gear like gloves, shovels, etc. Printing brochures, seed guides, and posters.
Equipment	\$ 1050	Binoculars, monitoring kits, Cameras for photo documentation, waterproof notebooks, and clipboards.
Outreach Materials	\$ 3000	Signage, workshop supplies, seed packets, stipends for guest educators and workshop facilitators.
Personnel	\$ 1100	Project lead stipend: \$800 Volunteer coordinator: \$300
Contingency	\$ 250	Plant replacement
Total:	\$ 9000	

Budget Justification:

\$800 will be for fuel for traveling between planting areas and outreach programs as well as for a onetime rental vehicle to carry larger equipment for planting. \$2000 will be for supplies like gloves, shovels, and watering cans for plant maintenance. Printing costs for posters and education materials for workshops. Also data collection materials like waterproof notebooks. The equipment costs listed will be for binoculars for pollinator monitoring, cameras for the same and transect materials like markers and signage posts. Outreach materials, \$3000, include signage, workshop supplies, seed packs, and stipends (\$2000) for guest educators or specialists to lead high-quality workshops. \$800 total stipend for project lead and \$300 stipend for volunteer coordinator. Then the contingency would be used for ongoing plant area maintenance. With partial funding I could limit the supplies and outreach materials by using rented equipment and switch to all online presented education material at workshops (powerpoint style). Stipends for guest educators could also be reduced or eliminated for a volunteer style approach.

References:

Allison, S. K. "When Is a Restoration Successful? Results from a 45-Year-Old Tallgrass Prairie Restoration." *Ecological Restoration*, vol. 20, no. 1, 1 Mar. 2002, pp. 10–17,

<https://doi.org/10.3368/er.20.1.10>. Accessed 12 Oct. 2019.

Harmon-Threatt, Alexandra N., and Stephen D. Hendrix. "Prairie Restorations and Bees: The Potential Ability of Seed Mixes to Foster Native Bee Communities." *Basic and Applied Ecology*, vol. 16, no. 1, Feb. 2015, pp. 64–72, <https://doi.org/10.1016/j.baec.2014.11.001>. Accessed 28 Jan. 2021.

"Prairie Restoration." *Prairie Nebraska*, 5 Jan. 2011, prairienebraska.org/restoration/. Accessed 7 Dec. 2025.

Ries, Leslie, et al. "Conservation Value of Roadside Prairie Restoration to Butterfly Communities." *Conservation Biology*, vol. 15, no. 2, Apr. 2001, pp. 401–411, <https://doi.org/10.1046/j.1523-1739.2001.015002401.x>.

Sexton, Aaron N., and Sarah M. Emery. "Grassland Restorations Improve Pollinator Communities: A Meta-Analysis." *Journal of Insect Conservation*, vol. 24, no. 4, 14 May 2020, pp. 719–726, <https://doi.org/10.1007/s10841-020-00247-x>.

Smith, Daryl. "The Tallgrass Prairie Center Guide to Prairie Restoration in the Upper Midwest." *Google Books*, 2025, books.google.com/books?hl=en&lr=&id=b_OinHbA7zgC&oi=fnd&pg=PR7&dq=prairie+restoration&ots=dA1bW0IJQl&sig=tbIR7RFZSoXVXsupaFDzqAeUOds#v=onepage&q=prairie%20restoration&f=false. Accessed 7 Dec. 2025.